NO CAP

UI / Design System Specification

Design it. Break it. Scale it. — for one learner, at zero cost.

DOCUMENT	**UI / Design System Specification**
VERSION	**v2.1 (personal app, zero-cost)**
DATE	**16 August 2026**
SCOPE	**Web/PWA primary, mobile read-only**
STATUS	**Working spec**

PRODUCT THESIS

A personal system-design workbench that turns passive reading into a repeatable loop of learning, visualization, reasoning, simulation, architecture practice, and interview readiness — with no paid dependencies and no accidental billing, on Cloudflare's genuine free tier, where core functionality remains usable at free-tier exhaustion.

What Changed: v2.0 to v2.1

The visual design system (color palette, glassmorphism, typography, motion) is **unchanged from v2.0**. The warm editorial palette and glass surfaces were already right. What changes in v2.1 is the **screen inventory and sequencing**: screens are tagged by tier (A/B/C/D), MVP screens are pruned to essentials, and optional screens (Voice, AI, Collab) are clearly marked as Tier C/B.

Preserved (Tier A screens)

- App shell (Smoke Glass sidebar + top bar + Cmd+K).
- Home / Today (Liquid Glass hero + Frosted cards).
- Daily Dose (without Voice + Cost micro-question).
- Roadmap (4 modes — Career view deferred to Tier B).
- Concept page (without Notes tab + Cost section).
- Interactive Lab (without cost/hr readout).
- Architecture Playground (solo + validation only).
- Review (without hands-free mode).
- Progress (5-dim mastery matrix).
- Search, Focus Mode, Glossary.

Resequenced (Tier B/C/D)

- Notes & Annotations → Tier B (v1.0).
- Collections & Study Lists → Tier B (v1.0).
- Code Sandbox → replaced by Code Walkthrough (Tier B, v1.0).
- Voice / Audio Mode → Tier C (v2.0, browser TTS).
- Cost Calculator → Tier B (v1.0, bundled pricing).
- Career Path Mapping → Tier B (v1.0).
- Collaborative Whiteboarding → Tier B (v2.0, Durable Objects).
- AI Mentor UI → Tier C (v2.0, quota-guarded).
- Calendar UI → Tier C (v2.0, optional).
- Mobile-specific screens → PWA-first; native mobile deferred.

> **DESIGN PHILOSOPHY (UNCHANGED)**
> NO CAP feels like a serious engineering cockpit with the warmth of a study workspace. Glassmorphism is functional, not decorative: frosted surfaces separate content layers without harsh borders, blur indicates depth and focus, and the warm palette signals craft over commodity. The product never looks like a generic AI template — no purple-to-pink gradients, no neon blue accents, no animated mesh backgrounds.

1. Design Direction

VISUAL NORTH STAR
Think: Linear x Vercel x Stripe Press, but with the warmth of a study workspace. A serious engineering cockpit with a chill builder personality. Modern, glass-rich, warm, technical, and never childish. Glass surfaces carry information; they are not chrome. Color carries meaning — amber for focus, forest green for mastery, rust for trade-off caution, deep red for failure.

NO CAP should feel like a serious engineering cockpit with the personality of a chill builder community. Avoid generic ed-tech visuals, excessive gradients, cartoon mascots, or "AI app" aesthetics. Use a restrained warm palette, crisp typography, layered depth via glass surfaces, technical diagrams, and playful (but occasional) microcopy. The product must look like it was designed by engineers who care about craft, not assembled from a generic SaaS template.

2. Color System

v2.0 palette preserved. The violet accent (AI-slop signal) is gone. The new palette is anchored on warm cream paper with deep charcoal ink, accented by burnt amber (focus, progress, selected state) and forest green (mastery, success, architecture cues). Every color carries meaning; no color is decorative.

Token	Role	Hex	Usage Rule
Paper / 0	Light canvas	#FAF7F2	Page background, light theme. Warm cream, not pure white.
Paper / 50	Subtle surface	#F2EDE3	Section dividers, secondary surfaces.
Card / 100	Card fill (light)	#ECE5D8	Frosted card under solid background. Use for non-glass cards.
Card / Glass	Glass card	rgba(255,255,255,0.55)	Frosted glass surface over imagery. Pair with backdrop-blur(12px).
Card / Glass Dark	Glass card (dark)	rgba(26,24,21,0.55)	Frosted glass on dark theme. Pair with backdrop-blur(16px).
Ink / 900	Primary text	#1A1815	Body text, headings. Warm near-black. Never use pure #000000.
Ink / 700	Body text	#2A2520	Long-form body. Slightly softer than Ink 900.
Ink / 500	Muted text	#6B6259	Captions, metadata, timestamps, secondary labels.
Border / 200	Hairline divider	#D6CFC0	Subtle dividers, table borders. 0.4-0.6px weight.

Token	Role	Hex	Usage Rule
Border / 300	Emphasis border	#A89B82	Selected card outline, focused input border.
Accent / Amber	Primary accent	#B45309	Focus, progress, selected state, primary CTAs. Sparingly.
Accent / Forest	Secondary accent	#166534	Mastery, success, architecture cues, completed states.
Accent / Rust	Tertiary accent	#7C2D12	Trade-off caution, "explain why" prompts. Used in callouts.
Warning	Caution	#92400E	Review due, trade-off caution, degradation notice.
Danger	Failure	#991B1B	Critical errors, security issues, broken architectures.
Success	Healthy	#166534	Mastered concepts, passing scores, healthy systems.
Info	Informational	#0F766E	Deep teal, used sparingly for "FYI" badges only. Never as primary.
Code / Ink	Code background	#1F1B16	Deep warm-black for code blocks.
Code / Parchment	Code text	#F5E6C8	Warm parchment text on code blocks. High contrast, low glare.

3. Glassmorphism System

Glassmorphism is the signature visual language of NO CAP. It serves three purposes: depth-layering (separating content planes without harsh borders), focus (blurring non-focused chrome to draw attention to active content), and craft (the product feels designed, not assembled). Glass is never used decoratively — every frosted surface must have a structural reason.

3.1 Glass Surface Tiers

Tier	Use Case	CSS Recipe	When to Use
Liquid Glass	Hero cards, focus surfaces	background: rgba(255,255,255,0.55); backdrop-filter: blur(24px) saturate(180%); border: 1px solid rgba(255,255,255,0.6); box-shadow: 0 8px 32px rgba(26,24,21,0.08), inset 0 1px 0 rgba(255,255,255,0.4);	Today card, active concept card, architecture canvas top bar. Maximum 2 per screen.
Frosted Glass	Standard cards, panels	background: rgba(255,255,255,0.45); backdrop-filter: blur(16px) saturate(160%); border: 1px solid rgba(214,207,192,0.6);	Concept cards, quiz option cards, lab control panels. Default card style.
Smoke Glass	Sidebars, navigation	background: rgba(250,247,242,0.7); backdrop-filter: blur(20px); border-right: 1px solid rgba(214,207,192,0.5);	Desktop sidebar, mobile bottom nav, sticky headers. Persistent chrome only.
Dark Glass	Code blocks, command palette	background: rgba(31,27,22,0.85); backdrop-filter: blur(20px); border: 1px solid rgba(245,230,200,0.1);	Code panels, command palette (Cmd+K), interview timer. Dark elements on light page.
Edge Glow	Focused/hover state	box-shadow: 0 0 0 1px #B45309, 0 0 24px rgba(180,83,9,0.15);	Active input, hovered card, selected architecture node. Replaces border highlight.

3.2 Glass Rules

· Glass surfaces require a backdrop with content or color behind them. Glass over flat white = wasted blur. Use solid cards instead.

· Backdrop-filter is not supported in all browsers (notably older Firefox). Fallback: solid Card / 100 background. Always provide a fallback.

- Maximum 2 Liquid Glass surfaces per screen. Too many glass tiers kills the depth effect and looks like a 2021 dribbble shot.
- Glass border opacity must be at least 0.5 — thinner borders disappear against blurred backgrounds.
- Text on glass must pass WCAG AA contrast. Test with the actual backdrop, not a solid color sample. If contrast fails, darken the glass background rather than the text.
- In dark mode, glass becomes translucent dark (rgba(26,24,21,0.55)) with light text. The blur stays the same; only the tint flips.
- Glass surfaces never animate their blur value. Animate opacity, transform, or border — never backdrop-filter (jank-prone).
- Reduced-motion mode: glass surfaces stay (they are structural), but the Edge Glow becomes a static 1px border in the accent color.

3.3 Liquid Glass Effect

Liquid Glass is the highest-tier glass surface, used for hero elements only. It combines a translucent fill, a strong backdrop blur, an inner highlight (top-edge light leak simulating refracted light), and a soft outer shadow. The effect should feel like a piece of polished sea glass resting on the page — not a frosted window.

> **IMPLEMENTATION NOTE**
> Liquid Glass uses an inset box-shadow on the top edge to simulate light refraction: `inset 0 1px 0 rgba(255,255,255,0.4)`. This is the secret ingredient that separates "real glass" from "frosted rectangle". Without it, the surface reads as flat. With it, the surface reads as dimensional.

4. Typography

Typography is the second pillar of the NO CAP visual identity. Noto Serif SC is the primary text face — it has the warmth and craft of an editorial serif while remaining highly legible at small sizes for technical content. SarasaMonoSC handles all code, ASCII diagrams, and numeric data with tabular figures.

Role	Font	Size	Weight	Leading	Notes
Display / Hero	Noto Serif SC	32-48px	Bold	1.15	Cover titles, page H1. Tight leading.
Section heading	Noto Serif SC	18-22px	Bold	1.3	H2 on content pages.
Subsection	Noto Serif SC	13-15px	SemiBold	1.4	H3, card titles, list headers.
Body	Noto Serif SC	14-15px	Regular	1.55	Long-form lesson text. 15px on web.
UI label	Noto Serif SC	12-13px	Medium	1.4	Buttons, nav items, tags.
Metadata	Noto Serif SC	11px	Regular	1.4	Timestamps, captions, hints. Muted color.
Code / data	SarasaMonoSC	13-14px	Regular	1.5	Code blocks, ASCII diagrams, IDs.
Numbers / metrics	SarasaMonoSC	14-32px	Regular	1.0	Tabular numerals (tnum). Always.
Microcopy	Noto Serif SC Italic	11-12px	Italic	1.4	"Let's cook." etc. Sparingly.

4.1 Numeric Typography

All numeric data — latency, throughput, cost estimates, mastery scores, page numbers, XP — uses SarasaMonoSC with `font-feature-settings: "tnum"` enabled. Tabular figures ensure numbers align vertically in tables and metrics cards, which is critical for the engineering-cockpit feel. Without tabular numerals, a column of "1.2ms, 12.4ms, 123ms" looks chaotic; with them, it reads as data.

5. App Shell

The desktop shell uses a Smoke Glass sidebar (persistent), a Smoke Glass top bar (sticky), and a content area that switches between solid cards (default) and Liquid Glass hero cards (Today, focused states). The shell is the frame; content is the painting. The frame stays calm. v2.1

prunes the sidebar to Tier A destinations only — Tier B destinations appear when their features ship.

```
+--------------------------------------------------------------------+
|  NO CAP        Search (Cmd+K)        Streak 17    Settings         |   <- Smoke Glass top bar
+------------+-------------------------------------------------------+
|            |                                                       |
|  Home      |                                                       |
|  Today     |                                                       |
|  Roadmap   |             MAIN CONTENT AREA                         |
|  Concepts  |                                                       |
|  Practice  |   (Liquid Glass for hero cards, Frosted Glass         |
|  Labs      |    for content cards, solid surfaces for              |
|  Playground|    long-form text)                                    |
|  Case Study|                                                       |
|  Interview |                                                       |
|  Review    |                                                       |
|  Progress  |                                                       |
|  Glossary  |                                                       |
|            |   --- Tier B destinations appear when shipped ---     |
|            |   Notes *      Collections *     Cost *               |
|            |   Career *     Code Walk *                            |
|            |                                                       |
|            |   --- Tier C destinations appear when shipped ---     |
|            |   Voice *      AI Mentor *                            |
|            |                                                       |
+------------+-------------------------------------------------------+
   ^-- Smoke Glass sidebar (persistent, blur(20px))
```

Desktop shell layout. * = appears when feature ships.

Desktop uses a persistent sidebar (Smoke Glass, 240px wide). Tablet collapses to a 64px icon rail with tooltips. Mobile uses a bottom navigation bar (Smoke Glass, blur(20px)) with five primary destinations: Home, Roadmap, Practice, Review, Profile. Less-frequent destinations live behind the Profile "More" sheet. **v2.1 prioritizes web/PWA — native mobile is deferred.** Mobile is "read and review" only initially (Daily Dose, Review, Flashcards, Progress, Audio); deep architecture work stays on desktop.

6. Home / Today Screen [Tier A]

The Home screen is the daily landing surface. It uses one Liquid Glass hero card (Today's Dose) plus a grid of Frosted Glass secondary cards. The visual hierarchy is unambiguous: hero commands attention, secondary cards invite exploration. v2.1 removes the Voice toggle (Tier C) and Cost micro-prompt (Tier B) from MVP — they reappear when their features ship.

- **Hero card (Liquid Glass):** "Today in 12 min" with current concept title, 1-line description, and a primary CTA button "Start". Amber accent on the duration chip.
- **Review queue card (Frosted Glass):** Due count, quick-start button, next-due timestamp. Forest green ring around the count when zero due.
- **Roadmap progress strip:** Horizontal phase chips with completion %. Current phase highlighted with Edge Glow.
- **Weak area insight card:** One recommended concept + one-line reason. "Your last 3 quiz attempts on Sharding averaged 58% - try this refresher."
- **Streak / momentum card:** Compact streak number, 7-day mini bar chart, recovery tokens remaining. No giant gamification wall.
- **Continue where you left off card:** Resumes the exact scroll position of the last lesson. Persists across devices via D1 sync.
- **Focus mode toggle (Tier A):** Small icon top-right of the page. Activates distraction-free surface with Pomodoro timer.
- **Real-world teardown card (Tier B, v1.0):** Rotates weekly. Architecture teardown teaser (Discord, Cloudflare, Netflix) with read-time chip.
- **Voice mode toggle (Tier C, v2.0):** Small icon in the hero card top-right. Tapping it converts the day's dose to audio narration via browser TTS.
- **Cost micro-prompt (Tier B, v1.0):** "Quick estimate: what's the dominant cost driver for an architecture like yesterday's?" Appears on alternate days.

7. Daily Dose Screen [Tier A]

```
+----------------------------------------------------+
|  [12 MIN]  [STANDARD]           [FOCUS]            |   <- Liquid Glass top bar
|                                                    |
|  TODAY'S CONCEPT                                   |
|  CAP Theorem                                       |
|  Why distributed systems force trade-offs.         |
|                                                    |
|  [ Start ]                                         |   <- Amber primary CTA
|                                                    |
|  01  Mental model          2 min    [done]         |
|  02  Visual                3 min    [current]      |
|  03  Interactive           3 min    [locked]       |
|  04  Quiz                  2 min    [locked]       |
|  05  Recall                2 min    [locked]       |
```

```
|                                                    |
|  [ Finish Today ]                                  |
+----------------------------------------------------+

  --- When Tier B/C features ship, the top bar gains: ---

+----------------------------------------------------+
|  [12 MIN]  [STANDARD]           [VOICE]  [FOCUS]  |
|  ...                                                |
|  06  Cost micro-question   1 min    [locked]  *    |   <- * Tier B (v1.0)
+----------------------------------------------------+
```

Daily Dose layout. Tier A MVP shown first; additions marked.

Each step is a Frosted Glass row in a vertical list. Completed steps show a forest green check; the current step shows an amber Edge Glow ring; locked steps are muted. Tapping Voice in the top bar (when shipped, Tier C) swaps the visual concept card for an audio player with chapter markers tied to the same five (now six) steps. Tapping Focus hides everything except the current step and a Pomodoro timer.

8. Roadmap UI [Tier A]

The roadmap is a map, not a checklist. v2.1 ships the four v1.0 modes (Tier A). Career View (v2.0 addition) is deferred to Tier B (v1.0) where it requires the Career Path feature.

Mode	Behavior	Tier
Guided Path	Recommended next step with prerequisite hints. Default view.	A
Explore Graph	Free navigation across concept dependencies. Phase-grouped clusters.	A
Mastery View	Color/shape state based on learn/recall/apply dimensions. Heat-map style.	A
Interview View	Highlights interview-relevant concepts only. Side panel shows frequency in real interviews.	A
Career View	Overlays role expectations (L3-L7, Staff, Principal) onto concepts. Shows gap analysis vs selected target role.	B

9. Concept Page [Tier A core, Tier B additions]

```
+-----------------------------------------------------+
|  Title              12 min        CORE             |
|  Cache Aside                                       |
|                                                    |
|  [Prerequisites]  [Related]  [Used in]  [Notes *]  |   <- * = Tier B Notes tab
|                                                    |
|  WHY IT EXISTS                                     |
|  ...                                               |
|                                                    |
|  HOW IT WORKS                                      |
|  [interactive flow - Liquid Glass frame]           |
|                                                    |
|  TRADE-OFFS                                        |
|  + fast reads        - stale data risk             |
|                                                    |
|  TRY IT                                            |
|  [ Open Cache Lab ]      [ View Code Walk * ]      |   <- * = Tier B Code Walkthrough
|                                                    |
|  COST IMPLICATIONS *                               |   <- * = Tier B section
|  At 10k QPS, this pattern adds ~$340/mo on AWS.    |
|                                                    |
|  CHECK YOURSELF                                    |
|  [ Question card ]                                 |
|                                                    |
|  COMMON MISTAKES                                   |
|  ...                                               |
|                                                    |
|  WHERE YOU SEE IT                                  |
```

```
|  Netflix / web apps / APIs                        |
|                                                   |
|  [ Mark understood ]  [ Add to review ]           |
|                      [ Add to collection * ]      |   <- * = Tier B Collections
+---------------------------------------------------+
```

Concept page layout. * = Tier B additions.

The concept page is the most important content surface in NO CAP. Tier A MVP ships with: title, prerequisites/related/used-in chips, WHY IT EXISTS, HOW IT WORKS, TRADE-OFFS, TRY IT (Open Lab), CHECK YOURSELF, COMMON MISTAKES, WHERE YOU SEE IT, and the Mark Understood / Add to Review actions. Tier B adds: Notes tab, Cost Implications section, View Code Walkthrough CTA, Add to Collection action.

10. Diagram and Visual Language

- Prefer flat architecture blocks with directional flows. No 3D, no skeuomorphic shadows on diagram nodes.
- Use consistent node categories and icons. Icons are monochrome (Ink / 700) — never colored.
- Synchronous arrows: solid line with arrowhead. Asynchronous events: dashed line. Data stores: cylinder shape. External actors: outlined rectangle with person icon.
- Annotate latency, throughput, or failure conditions directly on diagrams when relevant. Use SarasaMonoSC for the annotations so they read as data, not labels.
- Interactive diagrams include a legend (collapsible, top-right) and a reset button (icon-only, bottom-right).
- Use motion only to show causality: a request flowing from client to server to DB, a cache miss falling through to origin. Never animate static diagrams.
- Diagram backgrounds are Paper / 0 (light) or Code / Ink (dark). Never use a colored background to "make it pop".
- Node fill colors map to mastery state when shown in Mastery View: forest green (mastered), amber (in progress), rust (review due), muted gray (not started).
- Glass effect on diagram frames only when the diagram is the focal element of the screen. Embedded diagrams in lessons use solid Paper / 50 backgrounds.

11. Interactive Lab UI [Tier A]

```
+------------------------------+------------------------------+
|  SYSTEM VIEW                 |  CONTROLS                    |   <- Frosted Glass panels
|                              |                              |
|        [ LB ]                |  Requests/sec  [ 10k ]       |
|         /|\                  |  Servers        [ 3 ]        |
|        / | \                 |  Algorithm      [ RR v ]     |
|       /  |  \                |  Node #2        [ FAIL ]     |
|     [A] [B] [C]              |                              |
|                              |  --- Live Metrics ---        |
|                              |  p95 latency    120 ms       |
|                              |  error rate     0.8 %        |
|                              |  throughput    9.2k rps      |
+------------------------------+------------------------------+
|  WHAT CHANGED?                                              |   <- Liquid Glass callout
|  Node B failed. Traffic redistributed to A and C.           |
|  Estimated impact: 0 dropped requests, +18ms p95.           |
+-------------------------------------------------------------+

  --- When Tier B ships, controls panel adds: ---

|                              |  cost / hr      $ 4.20  *    |   <- * = Tier B (v1.0)
```

Interactive Lab layout. Cost annotation is Tier B.

v2.1 ships labs with latency, error rate, and throughput readouts (Tier A). The cost/hr readout (Tier B, v1.0) uses bundled pricing data — no live API calls, zero Worker cost. The "What Changed?" callout uses Liquid Glass to signal it is the focal explanation — it sits visually above the panels it explains.

12. Architecture Playground UI [Tier A core]

```
+--------+---------------------------------+----------------+
| Palette| Canvas                          | Inspector      |
| Client |                                 | Node: Database |
| CDN    |         [API]                   | Replicas: 3    |
| LB     |         ------->                | Read ratio:90% |
| Cache  |                         [DB]    | Region: us-east|
| Queue  |                                 |                |
| Worker |                                 | [Validate]     |
|        |                                 | [Hint]         |
|        |                                 | [Save]         |
|        |                                 | [Submit]       |
+--------+---------------------------------+----------------+

  --- Tier B (v1.0) additions: ---

  Inspector adds:  Cost/mo: $1240  (from bundled pricing)
  Footer adds:     Total architecture cost: $4,820/mo
                   [Export: PNG] [Mermaid] [Excalidraw] [Embed iframe]
  Top-right adds:  [Share...] [Version history]

  --- Tier B (v2.0) additions: ---

  Top-right adds:  [Invite to collab]
  Canvas shows:    Live cursors of other participants (colored, named)
```

Architecture Playground. Tier A MVP first; additions when shipped.

Tier A MVP: solo editing, deterministic validation, save as JSON graph in D1. Tier B (v1.0) adds per-node cost readouts (bundled pricing), total architecture cost footer, multi-format export, version history, and shareable URLs. Tier B (v2.0) adds collaborative editing via Durable Objects + WebSockets with live cursors and voice chat.

13. Case Study UI [Tier A core]

Case studies use a stepper. Each step unlocks progressively: Requirements → Estimation → APIs → Data → HLD → Deep Dive → Failure → Trade-offs. v2.1 defers the Cost step (Tier B, v1.0) and Interview Summary step (Tier B, v1.0) to when their features ship. At each stage, offer "I want to try first" and "Show explanation" buttons.

14. Interview Mode UI [Tier A text, Tier C voice]

Interview mode feels different from the rest of the product. Fewer decorative cards, more focused workspace. Visible timer (top-center, large tabular numerals), prompt pane (left, Frosted Glass), architecture canvas (center, solid background for focus), notes pane (right, plain text). The user can hide hints completely via a single "Focus" toggle. v2.1 ships text-mode interview (Tier A). Voice Interviewer (Tier C, v2.0) adds: AI interviewer asks questions aloud (browser TTS), learner speaks their answers (transcribed via browser STT), final rubric includes a communication-clarity score.

15. Review UI [Tier A core]

```
+--------------------------------------------------+
|  REVIEW DUE: 7                                   |    <- Tier A MVP
|                                                  |
|  What does CAP actually say?                     |
|                                                  |
|  A  You can always choose two of three.          |
|  B  Under partition, you cannot guarantee both   |
|     consistency and availability.                |
|  C  ...                                          |
|                                                  |
|  [ Reveal ]                                      |
|                                                  |
|  Confidence:                                     |
|  [ Forgot ]  [ Hard ]  [ Good ]  [ Easy ]        |
|                                                  |
|  Next review in: 4 days  (was: 1 day)            |
+--------------------------------------------------+

  --- Tier C (v2.0) addition: ---

|  REVIEW DUE: 7                 [Voice mode]  [*]  |    <- * browser TTS hands-free
```

Review card. Voice mode is Tier C.

16. Progress UI [Tier A core]

Widget	Purpose	Tier
Mastery matrix	5-dimension grid (Learn, Recall, Apply, Explain, Interview) across concept families. Heat-map style. v1.0 expands to 7 dims (+ Cost + Code).	A
Concept heatmap	Visualize weak and review-due areas across the full roadmap. Click any cell to drill into remediation.	A
Momentum chart	Weekly learning consistency. Bar chart, 7-day rolling. Forest green for above-target, amber for below.	A
Application score	Architecture and simulation performance trend. Line chart, last 30 attempts.	A
Interview readiness	Interview rubric trend across attempts. Radar chart across 5 dimensions (Communication, Breadth, Depth, Trade-offs, Estimation).	A
Notes synthesis	Personal knowledge graph rendered as a force-directed layout. Shows concept clusters the learner has written about most.	B
Cost accuracy	Cost reasoning accuracy over time. Shows estimate-vs-actual delta as a scatter plot.	B

Widget	Purpose	Tier
Career gap analysis	Radar chart comparing current mastery to target role expectations. Highlights top 3 gaps.	B
Quota indicator (subtle) [NEW]	Small healthy/warning indicator in top bar (not a dashboard). Normal: green dot "systems healthy". Near-limit: amber "AI temporarily limited". Limit: "AI Mentor back tomorrow". Raw numbers live in Settings → System Health.	A

v2.1 adds a **subtle quota indicator** in the top bar (Tier A) — not a prominent dashboard. The product should not feel like a Cloudflare dashboard. Normal state shows a small green dot with "systems healthy" tooltip. Near-limit shows an amber dot with "AI temporarily limited today". Limit-reached shows "AI Mentor is back tomorrow. Core NO CAP features remain fully available." Raw quota numbers (Worker requests, D1 rows, AI neurons) live under **Settings → System Health** for those who want the detail.

17. Tier B Screens (v1.0)

These screens ship in v1.0 alongside their corresponding features. They follow the same design system (warm palette, glass surfaces, tabular numerals).

17.1 Notes & Annotations [Tier B]

Personal notes attached to concepts. Three layers: (1) Margin annotations — short scribbles anchored to a specific paragraph or diagram element, shown as small amber dots in the margin that expand on hover. (2) Highlights — text selections colored with one of four highlight colors (amber, forest, rust, info teal), each mapped to a learner-defined category. (3) Concept notes — longer free-form notes attached to the concept as a whole, shown in a dedicated Notes tab on the concept page.

17.2 Collections & Study Lists [Tier B]

A grid of Frosted Glass collection cards. Each card shows: title, concept count, progress ring (forest green for completed, amber for in-progress), last-studied timestamp. Cards are drag-to-reorder. Tapping a card opens a focused study list view. System-defined starter collections: "Google L5 interview pack", "AWS SAA prep", "Foundations refresher", "Staff eng breadth pack". (No sharing in MVP — this is a personal app; sharing arrives in Tier B v2.0.)

17.3 Code Walkthrough Mode [Tier B] (replaces Code Sandbox)

```
+----------------------------------------------------+
|  pattern.py                                        |
|                                                    |
|  1   from redis import Redis                       |
|  2                                                 |
|  3   cache = Redis(...)                            |
|  4                                                 |
|  5   def get(k):                                   |
|  6       v = cache.get(k)                          |
|  7       if v:                                     |
|  8           return v                              |
|  9       v = db.fetch(k)                           |
|  10      cache.set(k, v, ttl)                      |
|  11      return v                                  |
|                                                    |
|  [Explain line]  [Modify]  [Predict output]        |
|  [Identify bottleneck]  [Map to architecture]      |
+----------------------------------------------------+
```

Code Walkthrough Mode. No execution — read, modify, predict only.

Code Walkthrough Mode replaces the v2.0 Code Sandbox. Instead of executing arbitrary code (which requires expensive sandboxed infrastructure), Code Walkthrough shows a reference implementation with interactive prompts: explain a line, modify the code and predict the output (learner reasons, then reveals answer), identify the bottleneck in this implementation, map this code to the architecture diagram. The learner can edit the code in the editor, but "running" it shows an authored expected output, not a live execution. This delivers 90% of the educational value at 0% of the infrastructure cost. Live execution (Tier D) can be added later if constraints change.

17.4 Cost Calculator [Tier B]

```
+----------------------------------------------------+
|  Architecture: WhatsApp-style messaging            |
|  Pricing dataset: AWS &mdash; bundled (as of 2026-08-01)  |
|                                                    |
|  +--------------------------------------------+    |
|  | Service          Qty   $/mo    % of total |    |
|  | ----------------------------------------- |    |
|  | EC2 (m6i.xlarge)   6   $1,420      34%    |    |
|  | RDS (db.r6.2xl)    2   $1,840      44%    |    |
|  | ElastiCache        3   $ 480       11%    |    |
|  | S3                 -   $ 120        3%    |    |
|  | Data transfer      -   $ 340        8%    |    |
|  | ----------------------------------------- |    |
|  | TOTAL                  $4,200             |    |
|  +--------------------------------------------+    |
|                                                    |
|  Cost per daily active user (10M DAU): $0.00042    |
|                                                    |
|  Pricing dataset updated: 2026-08-01 (quarterly)   |
|                                                    |
|  [Open in Playground]  [Save to architecture]      |
+----------------------------------------------------+
```

Cost Calculator with bundled pricing dataset.

The Cost Calculator uses a **bundled pricing dataset** (not live APIs). The dataset covers ~20 common services across AWS (Tier B), with GCP and Azure added later. It is versioned in the git repo and refreshed quarterly via a PR. Every estimate shows the "as of" date. This eliminates the live-API dependency entirely — the calculator works offline, costs zero Worker requests after the initial page load, and never surprises the learner with a billing prompt.

17.5 Career Path Mapping [Tier B]

```
+----------------------------------------------------+
|  Target role: [Staff Engineer v]    Gap: 23%       |
|                                                    |
|         +------ Databases ------+                  |
|         |   **********  78%     |  +2 concepts     |
|         +-----------------------+                  |
|         +------ Distributed ----+                  |
|         |   ******      62%     |  +4 concepts     |
|         +-----------------------+                  |
|         +------ Cost & Capacity +                  |
|         |   ***         35%     |  +6 concepts (!) |
|         +-----------------------+                  |
|         +------ Reliability ----+                  |
|         |   **********  91%     |  ready           |
|         +-----------------------+                  |
|                                                    |
|  Top 3 gaps:                                       |
|  1. Multi-region active-active (Distributed)       |
|  2. Capacity planning with seasonality (Cost)      |
|  3. Disaster recovery RTO/RPO trade-offs (Reliab.) |
|                                                    |
|  [Generate 4-week gap-closing plan]                |
+----------------------------------------------------+
```

Career Path Mapping with gap analysis.

Career Path Mapping shows the learner's current mastery against the expected profile for a target role (L3, L4, L5, Staff, Principal Engineer). A horizontal bar per concept family shows current mastery % and the number of concepts needed to close the gap. The top 3 most-impactful gaps are surfaced as a prioritized list. A "generate gap-closing plan" button creates a 4-week focused roadmap targeting those specific gaps.

18. Tier C Screens (v2.0, optional)

These screens ship in v2.0 as opt-in enhancements. They use browser-native APIs (Web Speech API for TTS/STT) and Cloudflare Workers AI (quota-guarded). The app works perfectly without them — they are never architectural dependencies.

18.1 Voice / Audio Mode [Tier C]

```
+-----------------------------------------------------+
|  <|  CAP Theorem              12:34 / 14:20  |>  |   <- Liquid Glass player
|                                                     |
|  waveform scrubber (SarasaMonoSC visual)            |
|                                                     |
|  Chapter 03 of 06 - Interactive                    |
|                                                     |
|  [x] 01 Mental model          2:10                  |
|  [x] 02 Visual                3:05                  |
|  [>] 03 Interactive           3:00                  |
|  [ ] 04 Quiz                  2:00                  |
|  [ ] 05 Recall                2:00                  |
|                                                     |
|  Speed: [0.8] [1.0] [1.25] [1.5] [2.0]              |
|  Voice:  [Browser default]                          |
|                                                     |
|  [ Hands-free flashcards ]                          |
+-----------------------------------------------------+
```

Voice mode player using browser-native TTS.

Voice Mode uses the **Web Speech API (SpeechSynthesis)** — browser-native, zero server cost, zero quota. Voice quality varies by browser/OS (Chrome on macOS is excellent; some Linux browsers are limited). The UI always shows the current voice and offers a "show text alongside" fallback. Hands-free flashcards mode lets users swipe through review cards with audio playback. **No paid TTS provider is used in v2.1.** External TTS (ElevenLabs-quality) is Tier D — deferred until constraints change.

18.2 AI Mentor [Tier C]

AI Mentor uses **Cloudflare Workers AI** with the free 10k-neurons/day allocation. A quota guard wraps every request: when daily neurons are exhausted, the mentor UI shows "AI quota reached for today - here is the authored explanation instead" and falls back to deterministic content. The learner sees current quota usage in the mentor panel. AI never mutates mastery scores; it can only suggest. AI output is labeled with a small amber "AI" chip. Sessions are stored in D1 and deletable.

18.3 Collaborative Whiteboarding [Tier B, v2.0]

```
+-----------------------------------------------------+
|  Architecture: WhatsApp-style     [Invite]  [End]  |
|                                                     |
|  Live: you (amber), alex (forest), sam (rust)       |
|                                                     |
|  +---------+    +---------+    +---------+          |
|  | Client  |-->|   LB    |-->|  API    |           |
```

```
|  +---------+   +---------+   +----+----+          |
|                                   |               |
|                              +----v----+          |
|                              |   DB    |          |
|                              +---------+          |
|                                                   |
|  Cursor: alex (forest) is editing the DB node     |
|  Comment: sam - "should we add a read replica?"   |
+---------------------------------------------------+
```

Collaborative whiteboarding via Durable Objects.

Collaborative whiteboarding uses **Durable Objects** for presence and CRDT sync. No Redis, no Yjs-server, no S3 snapshots — Durable Objects handle all of it on the Cloudflare free tier (100k requests/day, 13k GB-s/day). Max 4 concurrent editors for personal use. Sessions auto-end after 2h of inactivity. Snapshots save to R2 on session end.

19. Search

Global search opens via Cmd/Ctrl+K as a Dark Glass command palette overlaying the page with a 24px backdrop blur. Results are grouped by type (Concepts, Case Studies, Labs, Glossary) and show type, phase, difficulty, and mastery state for each row. Tier A MVP searches concepts and glossary only. Tier B (v1.0) adds Notes and Collections to the searchable corpus. Tier A uses client-side fuzzy search over static content — zero Worker requests for search. Tier B server-side search uses D1 FTS5 (SQLite full-text search, free).

20. Microcopy

Situation	Copy style
Start	"Let's cook." / "Start the dose." / "Open the lab."
Correct	"Yep. That's the trade-off." / "Locked in."
Wrong	"Close. The catch is..." / "Almost - what breaks at scale?"
Failure	"You just found the bottleneck." / "That's the SPOF."
Mastered	"Locked in." / "That concept is yours."
Review due	"Your brain wants a rematch." / "Time to dust this off."
Cost overshoot	"That's $2k more than the reference - what's driving it?"
Voice unavailable	"Browser TTS not available. Text version below."
AI quota reached	"AI quota reached for today. Authored explanation below."
Focus mode on	"Distractions off. Go deep."
Quota warning	"Heads up: 80% of today's Worker requests used. Consider batching."

Microcopy stays occasional. The product must never become a joke app; the content remains professional. Microcopy appears at most once per screen, never in body text, and never in error states that need precise information (use plain language there).

21. Accessibility

· Full keyboard navigation on web. Every interactive element is reachable via Tab and actionable via Enter/Space.

· All diagrams have text alternatives or adjacent explanation. Voice mode reads diagram alt-text aloud as authored descriptions, never auto-generated captions.

- Color is never the only way to communicate mastery or failure. Always pair color with shape, icon, or text label.
- Reduced-motion preference disables non-essential animation, including backdrop-blur transitions (replaced with instant opacity changes). Glass surfaces stay structural; their motion is removed.
- Minimum contrast meets WCAG AA. We target AAA wherever feasible. Text on glass surfaces is tested against the actual backdrop, not a solid sample.
- Interactive canvas (Architecture Playground) provides keyboard alternatives for core actions: Tab to navigate nodes, Enter to select, arrow keys to move, Delete to remove.
- Touch targets on mobile are at least 44x44pt. Bottom nav items are 60pt tall.
- Dyslexia-friendly font option: toggles body text to OpenDyslexic or a high-readability sans-serif (system fallback).
- Screen reader compatibility: all interactive elements have ARIA labels. Dynamic content updates use aria-live regions. Voice mode player controls have full keyboard support.
- High contrast mode: replaces glass surfaces with solid cards using maximum-contrast color pairs (preserves all information, removes blur).

22. Responsive Breakpoints

Breakpoint	Primary behavior
< 640px (mobile)	Single-column app, bottom nav (Smoke Glass). Read/review only: Daily Dose, Review, Flashcards, Progress, Audio. No architecture playground.
640-1024px (tablet)	Compact 64px icon rail sidebar, two-panel lesson/lab where space permits. Architecture Playground uses full width.
1024-1440px (desktop)	Full desktop shell with 240px Smoke Glass sidebar, content max-width 1100px, optional inspector pane. Primary surface for deep work.
> 1440px (wide desktop)	Wide canvas, content max-width 1280px (centered), optional inspector pane. Architecture Playground uses full available width.

v2.1 prioritizes web/PWA. The mobile experience is "read and review" only initially — Daily Dose, Review, Flashcards, Progress, Audio. Deep architecture work, case studies, and the playground stay on desktop. This is a deliberate scope cut: most of NO CAP's strongest features (architecture canvas, simulations, diagrams, case studies, dashboards) are desktop-heavy, and trying to make every screen identical on mobile would dilute the desktop experience. Native mobile app (React Native) is Tier D — deferred until the web/PWA is solid.

23. Motion

v2.1 keeps the v2.0 motion language. All motion uses spring physics (natural deceleration), never linear easing. Motion is reserved for causality and spatial transitions — never decoration. Reduced-motion users get the same information via instant state changes.

Context	Motion	Duration	Easing
Page transition	Cross-fade with subtle scale (0.98 -> 1.0)	180ms	spring (stiffness 300, damping 30)
Card hover	Lift 2px, Edge Glow appears	120ms	spring (stiffness 400, damping 28)
Concept graph focus	Dependencies fade in, unrelated nodes dim to 40%	300ms	spring (stiffness 200, damping 26)
Lab parameter change	Metric values count up/down to new value	240ms	spring (stiffness 250, damping 22)
Request flow animation	Animated dot travels along edge from client to server to DB	600ms per hop	linear (causality)
Architecture validation	Findings appear one-by-one with stagger	80ms per item	spring (stiffness 350, damping 30)

Context	Motion	Duration	Easing
Interview timer	Static, never animates. Tabular numerals.	0ms	none
Glass surface mount	Fade in (opacity 0 -> 1), no transform	160ms	ease-out
Reduced motion (any)	All animations become instant opacity changes	0ms	none

Avoid animated gradients, perpetual background motion, parallax effects, and any motion that does not communicate state change or causality. The product must feel calm — motion draws attention, and attention is the learner's scarcest resource.

24. Dark Mode

Dark mode is not "invert the colors". It uses depth layers (3 distinct dark surfaces) and warm-tinted blacks (never neutral zinc) to create a comfortable long-session reading environment. The amber accent reads as candlelight; the forest green reads as healthy signal. Both remain semantic.

Token	Light value	Dark value	Notes
Page background	#FAF7F2	#1A1815	Warm cream / warm near-black. Never pure white or pure black.
Surface / 50	#F2EDE3	#22201C	One step darker than page.
Card / 100	#ECE5D8	#2A2722	Solid card on dark. Warm.
Card / Glass	rgba(255,255,255,0.55)	rgba(42,39,34,0.55)	Frosted glass - flips tint, keeps blur.
Ink / 900	#1A1815	#F5E6C8	Primary text - becomes warm parchment in dark.
Ink / 700	#2A2520	#E8DCC4	Body text in dark mode.
Ink / 500	#6B6259	#9C8F7A	Muted text.
Border / 200	#D6CFC0	#3D352C	Hairline dividers.
Accent / Amber	#B45309	#D97706	Slightly brighter in dark for visibility.
Accent / Forest	#166534	#22C55E	Brighter forest green in dark mode.

25. Visual Quality Bar

QUALITY BAR
A learner should be able to look at any screen and answer: where am I, what am I learning, what should I do next, how am I doing, and why does this matter in a real system? If any of those questions is hard to answer, the screen is broken — regardless of how the glass or color looks.

26. UI Build Order

#	Step	Tier	Version
1	Design tokens + glassmorphism system + theme switcher + quota dashboard widget	A	0.1
2	App shell (Smoke Glass sidebar + top bar + Cmd+K)	A	0.1
3	Home / Daily Dose (Liquid Glass hero + Frosted cards)	A	0.1
4	Roadmap (4 modes: Guided, Explore, Mastery, Interview)	A	0.1
5	Concept lesson renderer	A	0.1
6	Quiz / review components	A	0.1
7	Progress dashboard (5-dim mastery matrix + quota widget)	A	0.1
8	Glossary + Focus Mode	A	0.1
9	Simulation lab shell (client-side, no cost/hr yet)	A	0.5
10	Architecture playground (solo + validation)	A	0.5
11	Case study stepper (8 steps, no Cost/Interview Summary yet)	A	0.5
12	Interview workspace (text mode only)	A	0.5
13	Notes & Annotations (margin + highlights + concept notes)	B	1.0
14	Collections & Study Lists (card grid + study list view)	B	1.0
15	Code Walkthrough Mode (replaces Code Sandbox)	B	1.0

#	Step	Tier	Version
16	Cost Calculator (bundled pricing dataset)	B	1.0
17	Career Path Mapping (skill tree + gap analysis)	B	1.0
18	Architecture playground extensions (cost + export + version history)	B	1.0
19	Case study extensions (Cost step + Interview Summary step)	B	1.0
20	Voice mode player (browser TTS + hands-free flashcards)	C	2.0
21	AI Mentor UI (Workers AI + quota guard + fallback)	C	2.0
22	Collaborative whiteboarding (Durable Objects + live cursors)	B	2.0
23	Voice interviewer (browser STT + TTS)	C	2.0
24	PWA service worker + offline lesson packs	B	1.0+
25	Mobile read/review optimization (PWA, not native)	A	0.1+